



TritonNav

Campus Navigation for UCSD First-Years & Transfers

Diego · UC San Diego · April 2026

SDUTC Application · Track 1: Developed Project

The Problem

"I was late to my first class because I couldn't find the building entrance."

— Every UCSD first-year and transfer student



Google Maps stops at the front door

No room-level directions. No building entrance guidance.



UCSD's website lists rooms, not routes

Students must cross-reference 3+ sites to find where to go.



No schedule-aware navigation exists

Every app requires you to know your building name first.

The Solution — TritonNav

One app. Login with TritonLink. Tap GO. You're there.



Schedule-Aware

Pulls your enrolled classes automatically from your TritonLink account



Room-Level Routing

Directions to the exact room — not just the building



One-Tap GO!

Single button to start turn-by-turn navigation from your GPS location



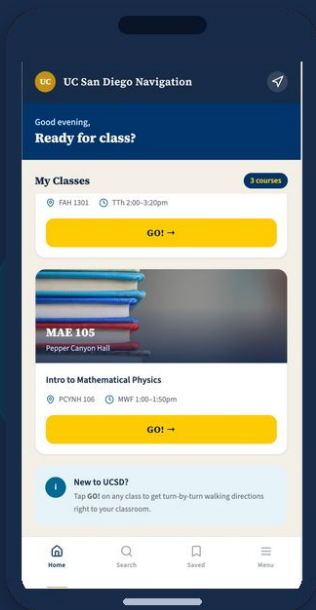
Campus Resources

Dining, events, food alerts, and Triton balance — surfaced in context

Live Prototype Screenshots

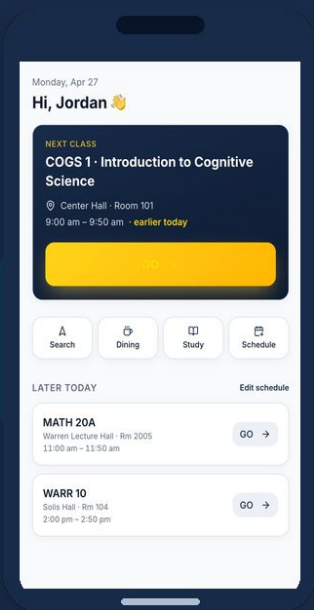
Figma • Lovable • Base44

Built with UCSD branding — navy, gold, Source Sans Pro typography



① My Classes Home

GO! button • Building photo •
Schedule times



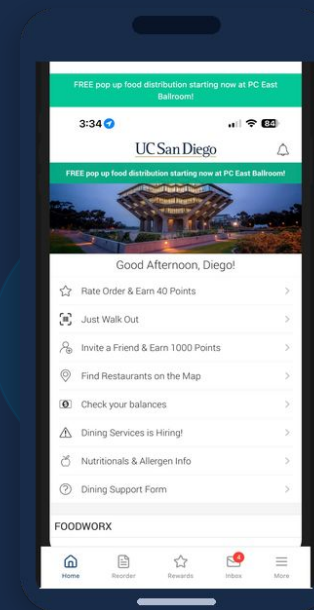
② Smart Schedule

Next class card • Quick-access
shortcuts



③ Live Navigation

Street view • Turn-by-turn • Room
directions



④ Dining & Resources

Food alerts • Balance • Dining hub

Technical Approach & Stack

Layer	Tool / Technology	Why It Fits
Frontend	React + Next.js	Mobile-first web app, fast routing, component reuse
Backend	Node.js + Express	Lightweight API server; proxies Google Maps key securely
Navigation	Google Maps Platform API	Directions, Maps JS, Places, Geocoding APIs
Authentication	TritonLink SSO	No new account needed — uses existing student credentials
Design	Figma · Lovable · Base44	Multi-platform prototyping; UCSD-branded across all tools
Deployment	Vercel + Render	Frontend on Vercel, Node backend on Render — both free tier

Current State vs. Roadmap

✓ Already Built

✓ Figma prototype — Version 22 (5 full screens)

✓ Lovable + Base44 interactive prototypes

✓ Next.js app shell + page routing

✓ SSO login screen (TritonLink-styled)

✓ My Classes home with course cards

✓ GO! navigation flow (mocked)

✓ Dining & resource hub screen

→ Next Milestones

→ Google Maps Directions API integration

→ Live GPS routing to real buildings

→ buildings.js — 30+ campus locations

→ TritonLink SSO live session management

→ User testing with incoming students

→ Dining API — live balance & menu data

→ IEEE paper + conference presentation

Summer Development Timeline

May 10

Application deadline

Submit SDUTC Track 1 application + Figma MVP

May–July

Google Maps API integration

Live routing, GPS, 30+ buildings in buildings.js

July 15

Checkpoint 1

Working nav for campus buildings + SSO live sessions

Aug 12

Checkpoint 2

User testing with first-years + dining API integration

Sept 4

Final paper + presentation

IEEE 2+ page paper + slide deck or demo poster

Final Deliverables & What I Need to Complete

Conference Deliverables (by Sept 4)

 IEEE 2+ page conference paper on TritonNav

 Slide deck OR demo/poster for conference

 Live working app with real Google Maps routing

 User testing results with first-year students

 2 progress checkpoints (July 15 + Aug 12)

What I Need to Complete the App

 Google Maps API key (Google Cloud Console)

 TritonLink SSO developer access / sandbox

 UCSD building coordinate dataset (or manual entry)

 Access to incoming first-year/transfer testers

 Deployment hosting (Vercel free + Render free)

Why TritonNav Matters

8,000+

New UCSD students
each year

~40%

Are transfers arriving
mid-year without orientation

Day 1

When TritonNav
is most needed

Why no existing app solves this:

✗ Google Maps — no room-level directions, no TritonLink integration

✗ UCSD Mobile app — general info only, no navigation or routing

✗ Wayfinder / campus maps — static PDFs, no personalization

✓ TritonNav — schedule-aware, room-level, SSO-integrated, built for students

TritonNav

Every Triton deserves to know where they're going — on Day 1.

I'm asking for:

- Feedback on scope and feasibility for summer timeline
- Guidance on TritonLink SSO developer access
- Connections to incoming first-years for user testing